

# Three AI systems agreed with each other. Not with us.

AI is increasingly being asked to make the same judgment call you make every session: figuring out why a student got something wrong. Before it shapes how your students are taught, we checked whether it can be trusted — and wanted to walk you through what we found.

Research briefing • **100** real tutoring conversations • **3** AI systems • **2** expert human reviewers

## BACKGROUND

# What is "learning analytics," and why do we care?

### IN PLAIN TERMS

Studying data about how students learn — including tutoring conversations — to see where they're stuck, and why. Tutors do this by ear already. AI can now do it at a scale no person could.

### WHY IT MATTERS HERE

Done well, that gets students the right help, faster. Done poorly — if the AI misreads the conversation — it points them toward the wrong help instead.

This briefing tests one question: **is the AI reading those conversations correctly?**

# One wrong answer, many possible reasons

## WHAT TUTORS ALREADY KNOW

- A wrong answer rarely means just one thing.
- It could be a guess, a mix-up, the wrong method, a shaky concept, or a small slip.
- Different reasons need different responses.

## TODAY'S QUESTION

- We asked 3 AI systems to diagnose 100 real wrong answers.
- We checked their answers against our own expert reviewers.
- If the AI systems agree with each other, are they right?

Not necessarily. Here's what we found.

# Six reasons a response can be wrong

| LABEL | WHAT IT MEANS                                                          | WHAT IT SOUNDS LIKE                                                     |
|-------|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| UNC   | Uncertainty — didn't really commit to an answer                        | <i>"Um, is it maybe 40?"</i>                                            |
| CTX   | Misattribution — reasoning imported from a different problem           | <i>Answers using a number from question 4, not question 5</i>           |
| COMP  | Operator selection — the wrong operation or approach for this question | <i>Adds when the problem calls for multiplying</i>                      |
| CON   | Conceptual gap — a wrong underlying idea                               | <i>Believes multiplying by 10 means "add a zero," even to a decimal</i> |
| PROC  | Procedural slip — right approach, execution mistake                    | <i>Correct method, arithmetic error partway through</i>                 |
| N/A   | Not diagnosable — can't be determined from the transcript              | <i>Inaudible, off-task, or too tangled to pin on one cause</i>          |

Every wrong answer got exactly one label. "Not diagnosable" isn't a sixth kind of mistake — it's an honest "we can't say."

# A fair test, not a demo

100 real wrong answers. 3 AI systems diagnosed why — seeing only what a tutor would see in the moment, not the correction after. Two expert reviewers did the same, then agreed on one answer per item to set the standard.

**100**

real tutoring conversations checked

**3**

AI systems compared, one per major vendor

**2**

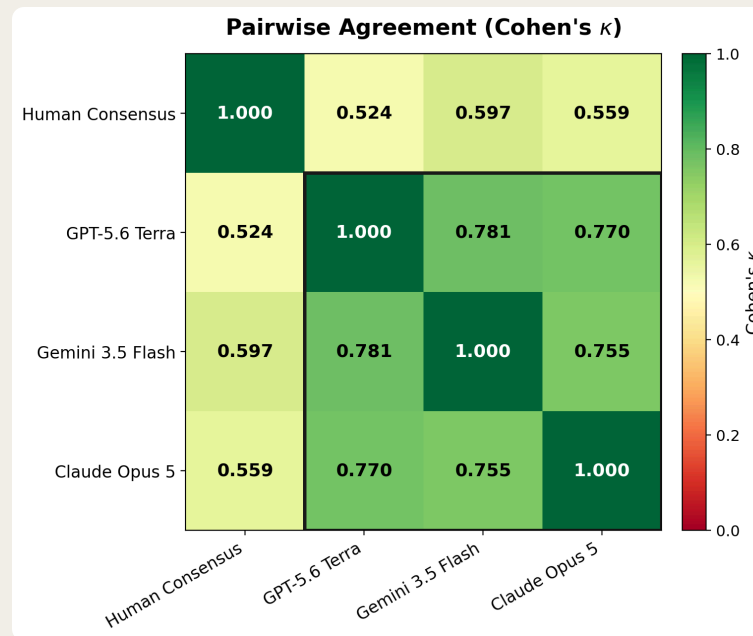
expert human reviewers, reconciled to one answer

OpenAI · GPT-5.6 Terra

Google · Gemini 3.5 Flash

Anthropic · Claude Opus 5

# The AI systems agreed with each other far more than they agreed with us



**How to read this:** each square scores how closely two raters agreed, 0 to 1 — greener means closer agreement. The outlined block, bottom right: the 3 AI systems agree with *each other* (.76–.78). The top row and left column: each AI system agrees with our reviewers less (.52–.60). Same systems, same data — just a different partner to agree with.

## WHAT ELSE WE FOUND

# The AI systems weren't just wrong — they were wrong the same way

- A small execution slip got mistaken for a deep conceptual gap **6 times out of 10** — the sharpest of these mix-ups.
- When unsure, the AI systems defaulted to the same guess: "**conceptual gap.**"
- Our reviewers said "**I can't tell**" 1 in 5 times. The AI systems said it about half as often — they'd rather guess.
- A stronger, slower AI didn't fix this. Results came out essentially the same.

WHAT THIS MEANS FOR YOU

# Using AI diagnosis without getting burned by it

- 01 **AI agreement isn't proof of being right.** It can mean a shared blind spot instead.
- 02 **Watch for over-use of "conceptual gap"** in AI-generated reports.
- 03 **Keep a person on the highest-stakes calls.**
- 04 **A system that says "I'm not sure" is more trustworthy** than one that always guesses.

QUESTIONS WELCOME

# Let's talk about what comes next

Not to scrap AI-assisted diagnosis — to use it in a way that earns your trust. Tell us what matches what you're seeing, and where human review matters most.

**A question for you:** Do these particular model errors matter instructionally — and if so, how? For example, does mistaking a procedural slip for a conceptual gap actually change what help a student would get?